Statistical Learning and Machine Learning

Exam Topics - January 2024

TOPIC 1

Consider a training data set of inputs $\{x_1 = 0.5, x_2 = 2.5, x_3 = 4.5\}$ and the corresponding targets or outputs $\{y_1 = 0.4, y_2 = 0.9, y_3 = 0.25\}$, as shown in Fig. 1 (left-sided figure below) in the 2D space. In this problem, it is required to fit an appropriate linear regression model to the above data using the least squares approach and then using the obtained model to predict the values y_i for given test data points x_i.

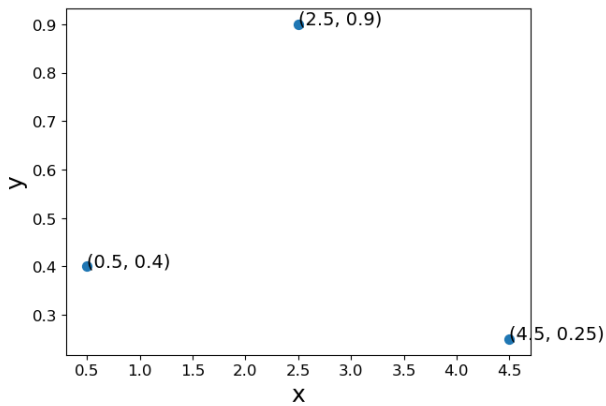

Fig. 1: The training set consisting of $N = 3$ data points (represented as blue circles) that are plotted in the 2D space.

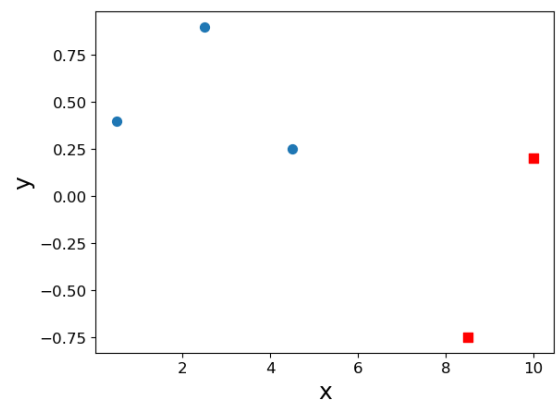

Fig. 2: The training set containing additional points (red circles).

1) (Score: 30 %) For the following choice for the basis functions $\boldsymbol{\phi}(x) = \{\phi_0(x), \phi_1(x), \phi_2(x)\} = \{1, x, x^2\}$ leading to the regression model $\hat{y}(x, w) = w_0 + w_1 x + w_2 x^2$, setup and write the *design matrix* $\boldsymbol{\Phi}$ and use it to write the *normal equations* for the least squares problem.
2) (Score: 15 %) Find the solution of the normal equations, obtained in the previous step, to calculate the weights (w_0, w_1, and w_2) of the regression model.
3) (Score: 25 %) Using the regression model obtained above, predict the values y_i for the training points $\{x_1 = 0.5, x_2 = 2.5, x_3 = 4.5\}$ as well as the test points $x_4 = 2$, and for $x_5 = 3.5$. Plot the corresponding points (x_i, y_i) in the 2D space in Fig. 1. Roughly draw the regression curve by joining those five points in Fig. 1.
4) (Score: 30 %) Consider Fig. 2 (right-sided figure above) where two more data points (red-squares) have been added to the original data set from Fig. 1 (blue-circles). Assume that the original model $\hat{y}(x, w) = w_0 + w_1 x + w_2 x^2$ (from part 1 of this topic) when trained on this new data does not produce acceptably accurate results. What changes would you make to the regression model to make it work better on this new data, and why? Note that you are not required to write or solve the normal equations in this case.

Appendix (Inverse of a matrix): It is highly recommended to use a software program to compute the **inverse** of a matrix. However, if you choose to do so manually, you can use the following equation for a 3×3 nonsingular matrix:

Let $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$, then: $A^{-1} = \frac{1}{|A|} \begin{bmatrix} (ei - fh) & (ch - bi) & (bf - ce) \\ (fg - di) & (ai - cg) & (cd - af) \\ (dh - eg) & (bg - ah) & (ae - bd) \end{bmatrix}$

where $|A| = \det(A) = a(ei - fh) - b(di - fg) + c(dh - eg)$ is the determinant of A.

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TOPIC 2

Consider a data set, represented by $\mathbf{X}$, containing $N = 6$ *training* data points shown by blue-square and red-circle markers in Fig. 3.

$$\mathbf{X} = \begin{bmatrix} 2 & 3 & 4 & 6 & 7 & 8 \\ 3 & 4 & 2 & 6 & 5 & 7 \end{bmatrix} \tag{1}$$

The blue-squared data points belong to class $\mathcal{C}_1$, with the corresponding target value $t = 0$, whereas the red-circled points belong to $\mathcal{C}_2$, with the target values $t = 1$. The corresponding target vector becomes $\mathbf{t} = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix}$. Using this training data, you are required to apply different classification algorithms to classify *test data points*, T1=$\begin{bmatrix} 5 \\ 4 \end{bmatrix}$ and T2=$\begin{bmatrix} 6 \\ 4.5 \end{bmatrix}$, that are shown as green triangles in Fig. 3, to either $\mathcal{C}_1$ or $\mathcal{C}_2$.

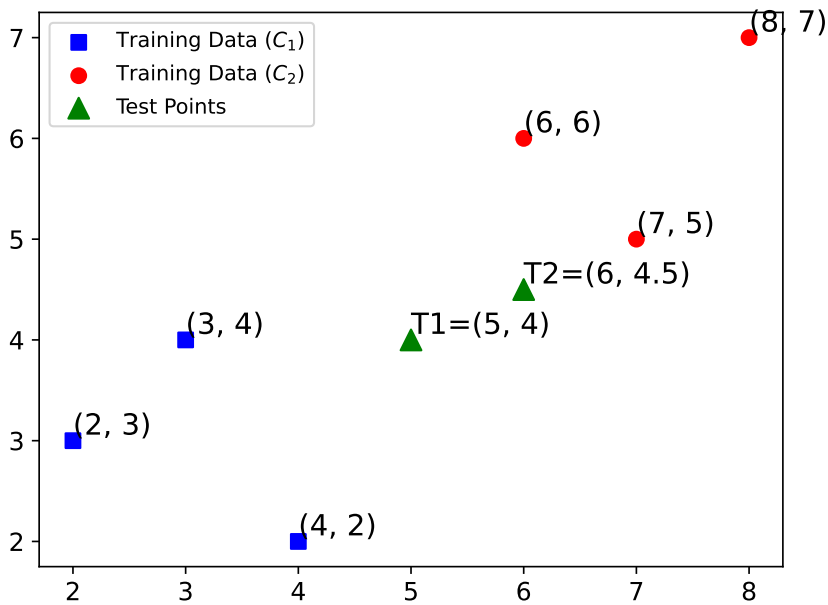

Fig. 3: The graphical illustration of the data set (both training and test sets) used in Topic 2.

1) (Score: 25 %) Apply the *K-nearest neighbour classification* algorithm to classify the two test data points to either $\mathcal{C}_1$ or $\mathcal{C}_2$. Use the value of the parameter $K = 1$.
2) (Score: 25 %) Apply the *nearest-centeroid-based* classification algorithm to classify the two test data points to either $\mathcal{C}_1$ or $\mathcal{C}_2$.

Apply *Fisher discriminant analysis* to:

3) (Score: 25 %) Calculate the projection vector $\mathbf{w} \in \mathbb{R}^2$ using the training data points (blue-squares and red circles in Fig. 3).
4) (Score: 25 %) Project the training data points to 1-dimensional space $\mathbb{R}$ using the projection vector $\mathbf{w}$, yielding $\{y_k\}_{k=1}^{N}$. Choose an *empirical* threshold value y_0 that maximizes the margin between the two classes on the projected training set. Subsequently, using that threshold value, classify the two test data points (green triangles in Fig. 3) to either $\mathcal{C}_1$ or $\mathcal{C}_2$.

Appendix (Inverse of a matrix): It is highly recommended to use a software program to compute the **inverse** of a matrix. However, if you choose to do so manually, you can use the following equation for a 2×2 nonsingular matrix:

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then: $A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

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TOPIC 3

Consider the single hidden-layer neural network for regression in Fig. 4, where x_i are the nodes in the input layer, $z_{1,i}$ are the nodes of the hidden layer, and y is the single node in the output layer. $\mathbf{w}_1$ and $\mathbf{w}_2$ are the weights (including biases) connecting the input layer x_i to the hidden layer $z_{1,i}$ and the hidden layer to the output layer y, respectively. The weights of the neural network are initialized to be:

$$\mathbf{w}_1 = \begin{bmatrix} w_{0,1}^{(1)} & w_{0,2}^{(1)} & w_{0,3}^{(1)} \\ w_{1,1}^{(1)} & w_{1,2}^{(1)} & w_{1,3}^{(1)} \\ w_{2,1}^{(1)} & w_{2,2}^{(1)} & w_{2,3}^{(1)} \end{bmatrix} = \begin{bmatrix} 0.2 & -0.8 & 0.6 \\ 0.7 & -1.0 & 0.3 \\ -0.5 & -0.3 & -0.4 \end{bmatrix} \qquad \mathbf{w}_2 = \begin{bmatrix} w_{0,1}^{(2)} \\ w_{1,1}^{(2)} \\ w_{2,1}^{(2)} \\ w_{3,1}^{(2)} \end{bmatrix} = \begin{bmatrix} 0.7 \\ 0.2 \\ -0.1 \\ -0.4 \end{bmatrix}$$

where $w_{i,j}^{(l)}$ denotes the weight connecting the ith node of the $(l-1)$th layer to the jth node of the lth layer.

A logistic sigmoid activation function $z_{1,i} = h_{ls}(a_{1,i}) = \frac{1}{1+exp(-a_{1,i})}$ is used in each of the nodes of the hidden layer, while an identity activation function $y = h_I(a_{2,1}) = a_{2,1}$ is applied to the output layer, where $a_{l,i}$ is the activation of the ith node of the lth layer and is given by a weighted sum of the inputs to the node.

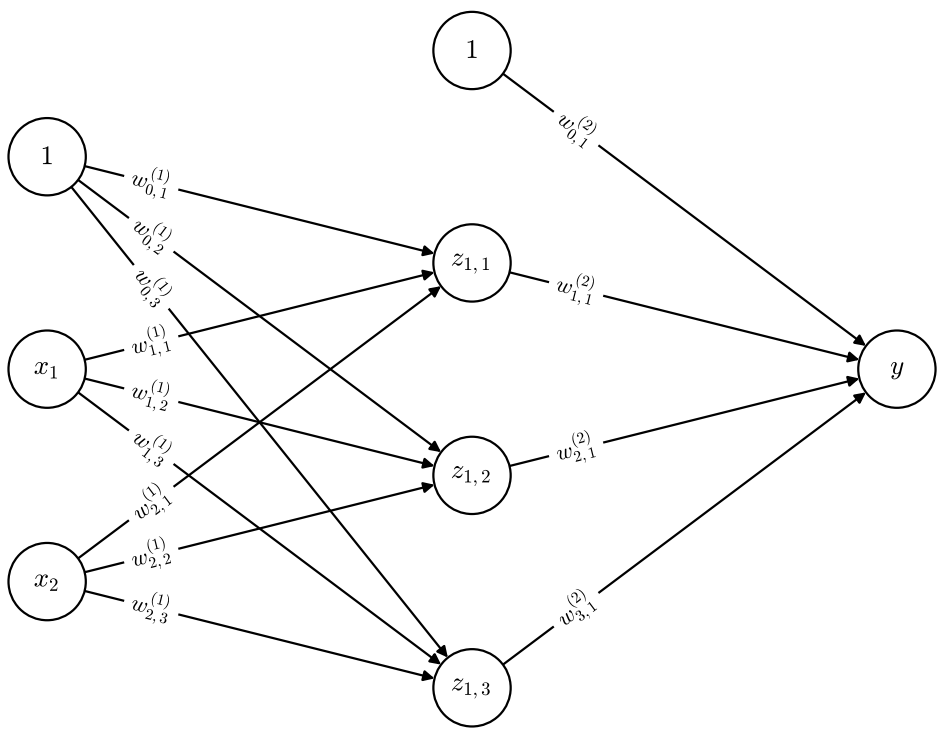

Fig. 4: A single hidden-layer neural network for regression.

1) (Score: 30%) Given the first data point $\mathbf{x}_1 = [0.6, -0.3]^T$ from the training set, what is the output value of the neural network? Show your intermediate calculations.
2) (Score: 20%) Assuming the output of the network is $y_1 = 0.531$ given the data point $\mathbf{x}_1$. What is the sum-of-squares error $E(t_1, y_1)$, if the corresponding target is $t_1 = -0.5$.
3) (Score: 30%) Given the data point $\mathbf{x}_1$ and the target t_1, use backpropagation to calculate the following partial derivatives of the error: $\frac{\partial E(t_1,y_1)}{\partial a_{2,1}}$, $\frac{\partial E(t_1,y_1)}{\partial w_{2,1}^{(2)}}$, $\frac{\partial E(t_1,y_1)}{\partial a_{1,2}}$ and $\frac{\partial E(t_1,y_1)}{\partial w_{1,2}^{(1)}}$. Show your intermediate calculations.
4) (Score: 20%) Assume, that the network has been trained on N training points $\{\mathbf{x}_n = [x_1, x_2]^T\}_{n=1}^N$, where $-1 \leq x_i \leq 1$ for $i \in 1, 2$. If the network is now presented with a new test data point $\mathbf{x}_{N+1} = [3, 5]^T$, can you trust the network's output/prediction for this new point? Explain why/why not?

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TOPIC 4

Consider the classification problem in Fig. 5 consisting of *training data points* from two classes $\mathcal{C}_0$ and $\mathcal{C}_1$ (red circles ● and blue squares ■, respectively) as well as a *test data point* ($\times$) with an unknown class. It is assumed that a *linear support vector machine (SVM)*, corresponding to the feature vector $\boldsymbol{\phi}(\mathbf{x}) = \mathbf{x}$, and with *no slack variables* has been trained to solve the classification problem.

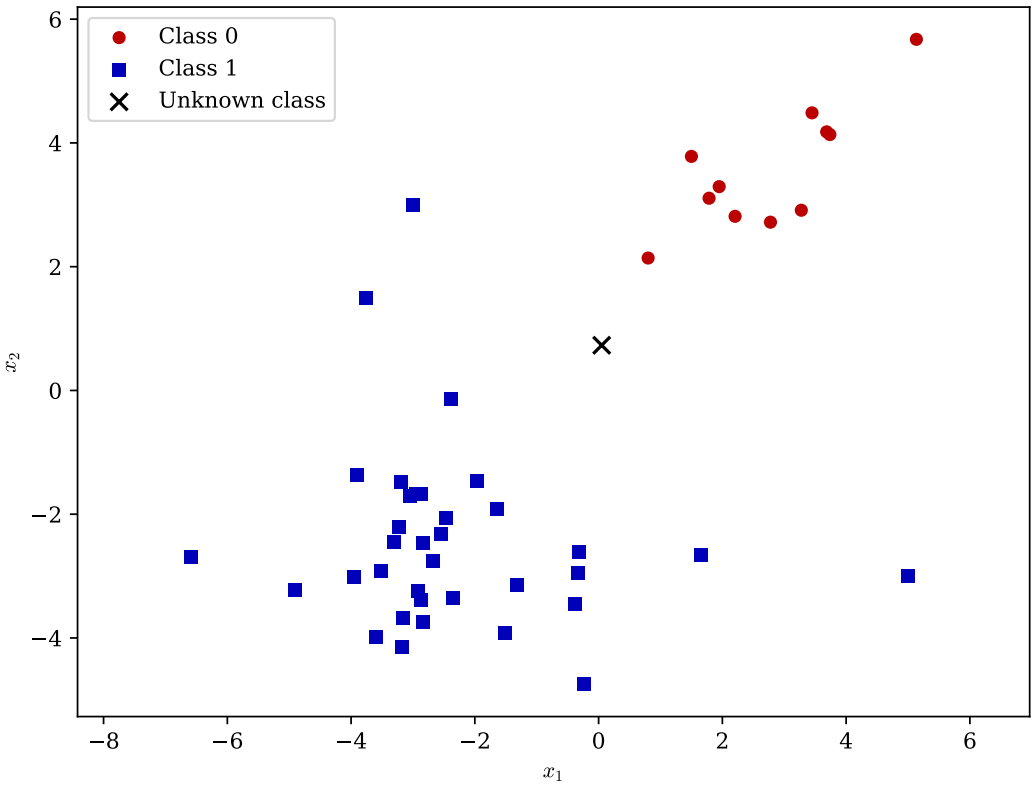

Fig. 5: Binary classification problem of the two classes $\mathcal{C}_0$ and $\mathcal{C}_1$ (red circles and blue squares, respectively) as well as a test data point ($\times$) with an unknown class.

1) (Score: 35%) Using your conceptual understanding of the SVM (*without* solving the actual SVM optimization problem), indicate which training data points may reasonably act as support vectors in Fig. 5 and also draw the possible decision function/boundary and margins of (linear) SVM. *Explain your reasoning*. Please use Fig. 5 for identifying the support vectors and drawing the decision boundary and margins.
2) (Score: 25%) Given a *test data point* indicated by the symbol ($\times$) in Fig. 5, use the SVM decision boundary from the previous question to assign this data point to either class $\mathcal{C}_0$ or $\mathcal{C}_1$.
3) (Score: 40%) Consider the two cases shown in Fig. 6 and Fig. 7, where some new data points have been added to the training set (shown by green triangles ► added to $\mathcal{C}_0$, and green diamonds ♦ added to $\mathcal{C}_1$, respectively). Considering that the *linear* SVM is again trained for binary classification in both cases, specify for each case whether the original decision boundary obtained earlier (in part 1 of the problem) will change or not. *Briefly explain your reasoning*. If the decision boundary changes, roughly draw the new decision boundary and margins and also identify the support vectors in Fig. 6 or Fig. 7 or both.

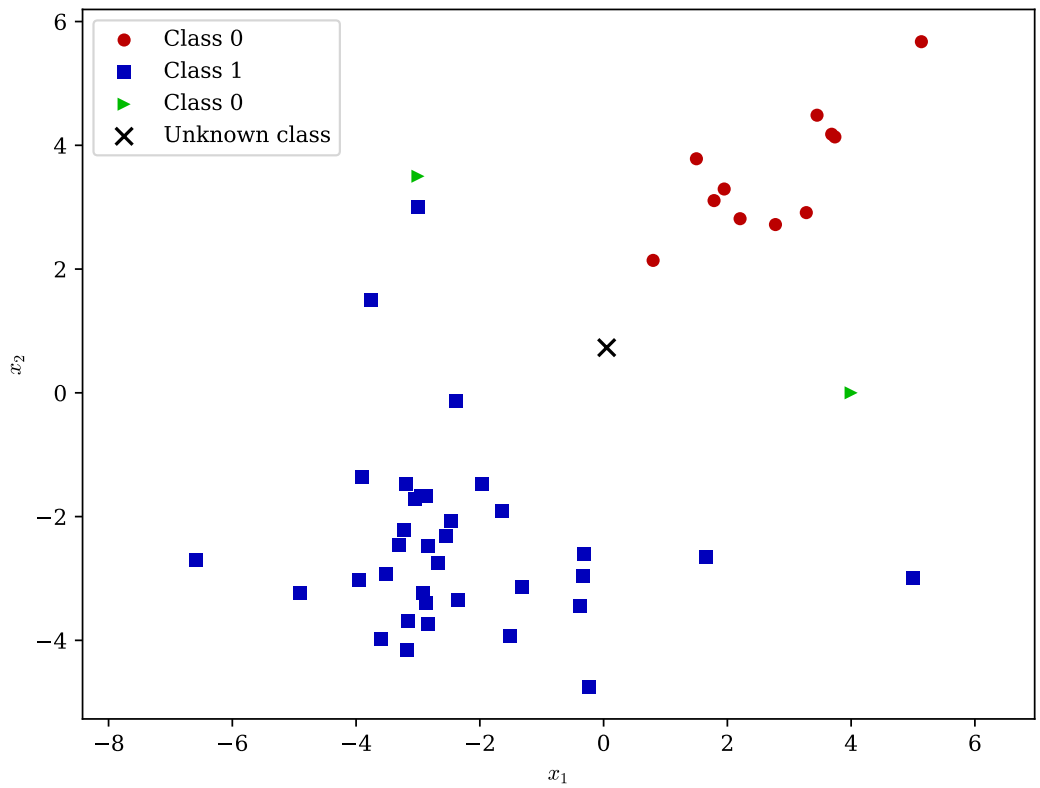

Fig. 6: Two new data points (green triangles) have been added to the training set for $\mathcal{C}_0$ (red circles).

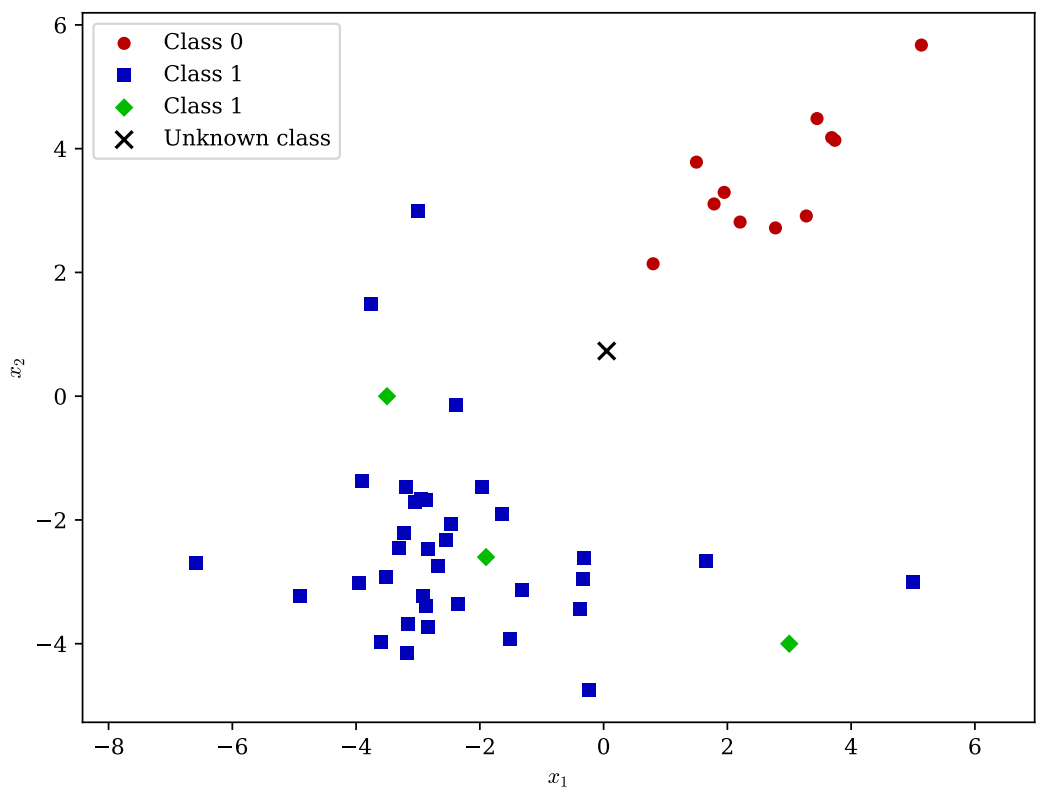

Fig. 7: Three new data points (green diamonds) have been added to the training set for $\mathcal{C}_1$ (blue squares).

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